9 ADVANCED LAYERING

Lesson overview

In this lesson, you'll learn how to do the following:

- Import a layer from another file.
- Clip a layer.
- Create and edit an adjustment layer.
- Use Vanishing Point 3D effects with layers.
- Create layer comps to showcase your work.
- Manage layers.
- Flatten a layered image.
- Merge and stamp layers.



This lesson will take less than an hour to complete. Copy the Lesson09 folder onto your hard drive if you haven't already done so. As you work on this lesson, you'll preserve the start files. If you need to restore the start files, copy them from the *Adobe Photoshop CS5 Classroom in a Book* DVD.



Once you've learned basic layer techniques, you can create more complex effects in your artwork using layer masks, adjustment layers, filters, and more layer styles. You can also add layers from other documents.

Getting started

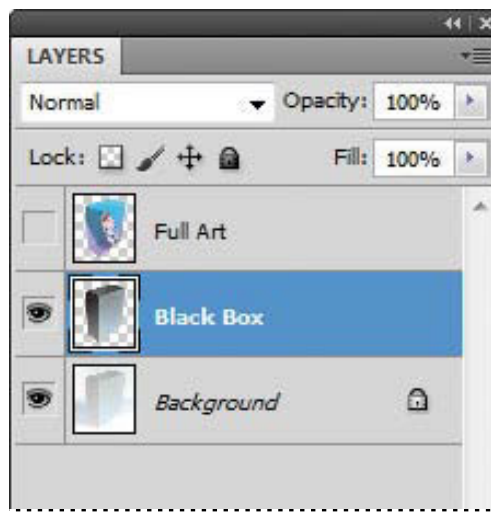
In this lesson, you'll combine several images to create a cell phone package. You'll create three different, multilayered designs, which you can display selectively using layer comps. You'll gain more experience with adjustment layers, layer effects, layer masks, and layer filters. Beyond this lesson, the best way to learn how to work with layers is by experimenting with combinations of filters, effects, layer masks, and layer properties in new ways.

1. Start Photoshop, and then immediately hold down Ctrl+Alt+Shift (Windows) or Command+Option+Shift (Mac OS) to restore the default preferences. (See "Restoring default preferences" on page 5.)
2. When prompted, click Yes to delete the Adobe Photoshop Settings file.
3. Click the Launch Bridge button (Br) in the Application bar to open Adobe Bridge.
4. In the Favorites panel in the upper-left corner of Bridge, click the Lessons folder, and then double-click the Lesson09 folder in the Content panel.
5. Study the 09End.psd file in the Content panel. If necessary, drag the thumbnail slider at the bottom of the window to enlarge the thumbnail.

Your goal in this lesson is to create a package prototype by assembling artwork from various files, layering the artwork, adding perspective, and then refining the design. You'll create several layer comps to show the design to your client.

6. Double-click the 09Start.psd file to open it in Photoshop. Choose File > Save As, rename the file **09Working.psd**, and click Save. Click OK in the Photoshop Format Options dialog box.
7. Drag the Layers panel by its tab to the top of the work area. Drag its corner to expand the panel so that you'll be able to see about 10 layers without scrolling.





The panel lists three layers, two of which are visible—the gray three-dimensional box displayed in the image window and the background stacked underneath it. The Full Art layer is hidden.

8 In the Layers panel, select the Full Art layer. Notice that even though the layer is selected, it remains hidden.

Clipping a layer to a shape

To start building a composite image, you'll add more artwork and clip it to a shape.

1 Click the Mini Bridge button (Mb) in the Application bar.

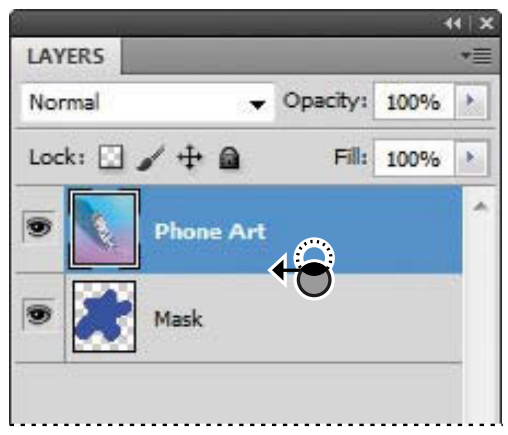
2 In the Mini Bridge panel, click Favorites, and then double-click the Lessons folder and the the Lesson09 folder in the Content pane. Finally, double-click the Phone_Art.psd file to open it.



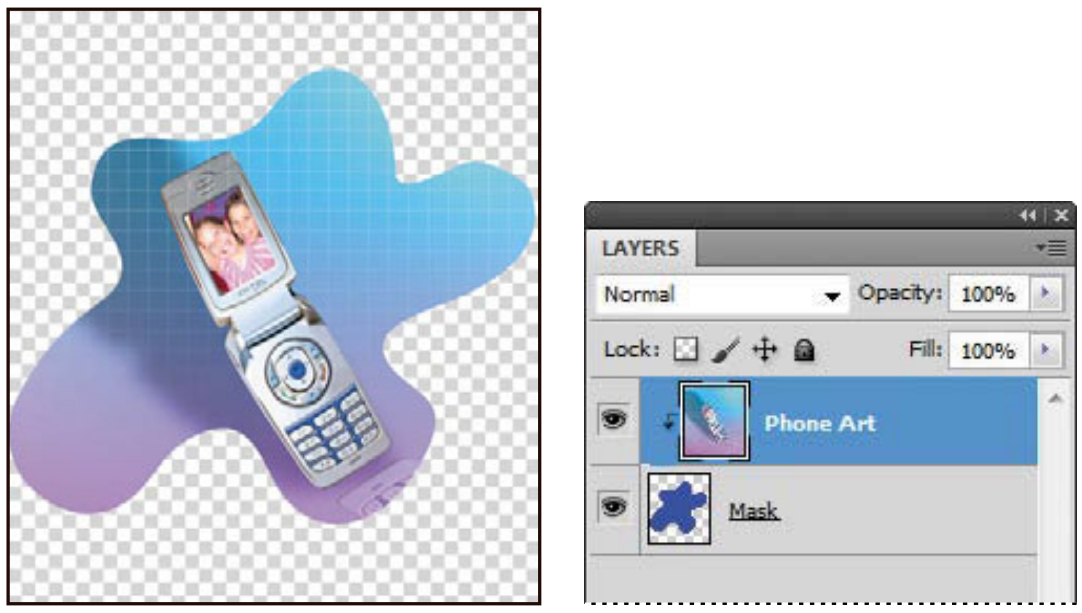
This file has two layers: Phone Art and Mask. You will clip the phone art image so that it fits within the freeform shape in the Mask layer below it.

3 In the Layers panel, make sure that the Mask layer is below the Phone Art layer. A clipping shape must be below the image that you're clipping.

4 Select the Phone Art layer. Then press Alt (Windows) or Option (Mac OS) as you position the pointer between the Phone Art layer and the Mask layer until it becomes a double-circle icon (), and click.

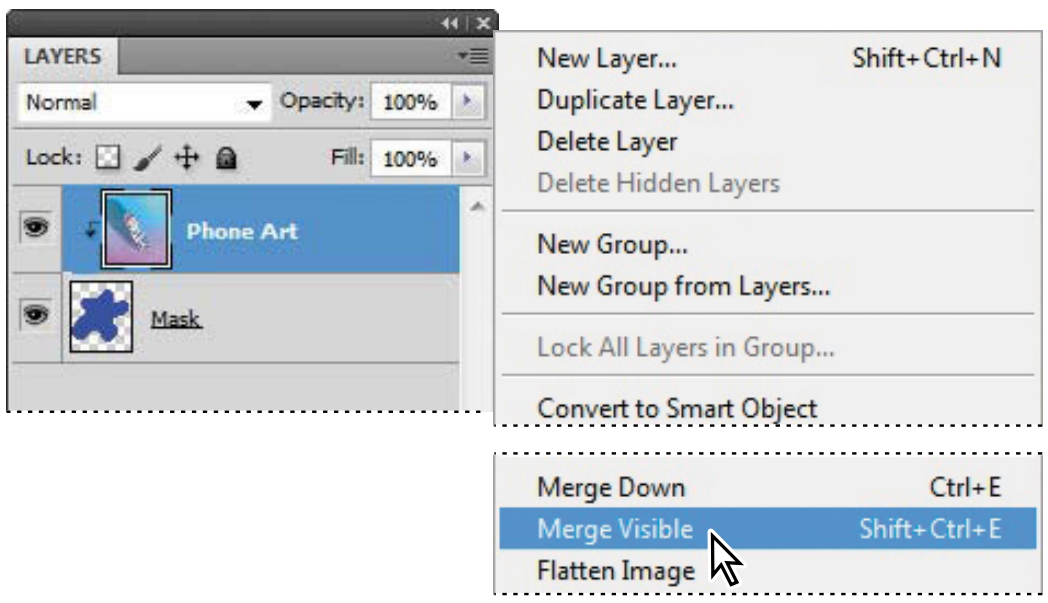


The thumbnail of the clipped layer, Phone Art, is indented in the Layers panel, and an arrow points to the layer beneath it, which is now underlined.



You will import this new image into the Start file. But first, you need to flatten the image to one layer.

5 Choose Merge Visible from the Layers panel menu.



You can merge the layers in other ways. But don't choose Flatten Image, because it would remove the transparency already set up in the file.

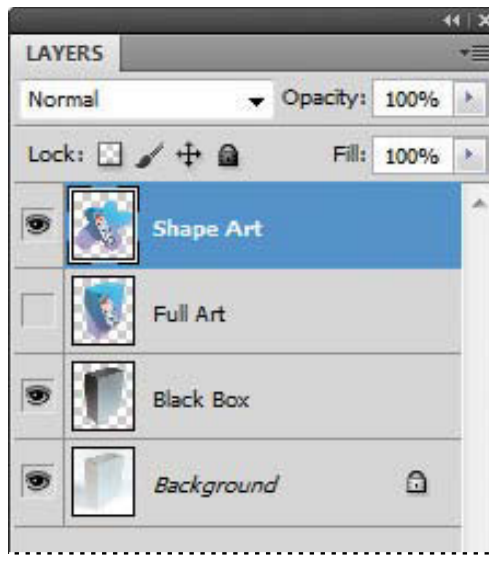
Now you'll add artwork from another file, simply by dragging and dropping.

6 Click the Arrange Documents button in the Application bar, and select a 2 Up layout so you can see both images.

7 With the Move tool, drag the merged Phone Art layer from the Layers panel into the 09Working.psd image window. The layer appears above the active layer (Full Art) and at the top of the 09Working.psd Layers panel. The artwork covers the box.

8 In the 09Working.psd Layers panel, double-click the Phone Art layer name, and type **Shape Art** to rename it. Press Enter or Return.





9 Choose File > Save to save your work so far.

10 Close the Phone_Art.psd file without saving your changes.

Setting up a Vanishing Point grid

The artwork you've added sits on top of the box—not exactly the effect you want. You'll fix that by making the artwork appear in perspective, wrapped around the box.

1 With the Shape Art layer selected in the Layers panel, press Ctrl+A (Windows) or Command+A (Mac OS) to select all of the layer's contents.

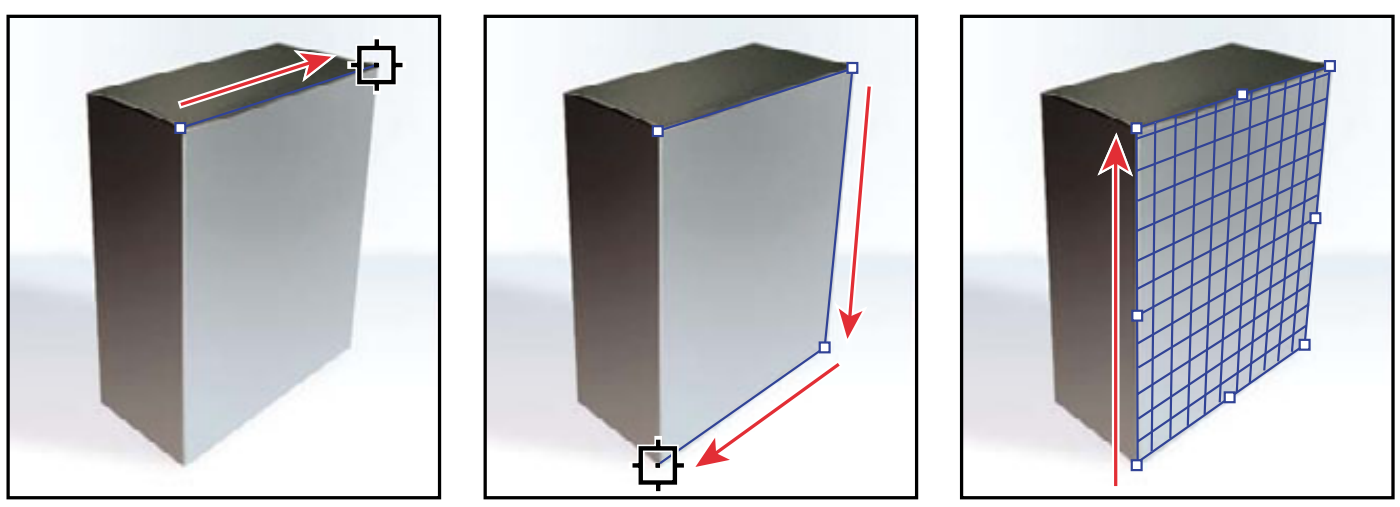
2 Press Ctrl+X (Windows) or Command+X (Mac OS) to cut the contents to the clipboard. Now only the box is visible, not the artwork.

3 Choose Filter > Vanishing Point. You'll use the Vanishing Point filter to draw a perspective plane that matches the dimensions of the box.

4 With the Create Plane tool ( ), which is selected by default, click the upper-left corner of the front of the box to begin defining the plane. It's easiest to define planes when you can use a rectangular object in the image as a guide.



5 Continue drawing the plane by clicking each corner of the box side. Click the last corner to complete the plane. When you complete the plane, a grid appears on the front of the box, and the Edit Plane tool ( ) is automatically selected. You can adjust the size of this grid at the top of the dialog box using the Edit Plane tool.

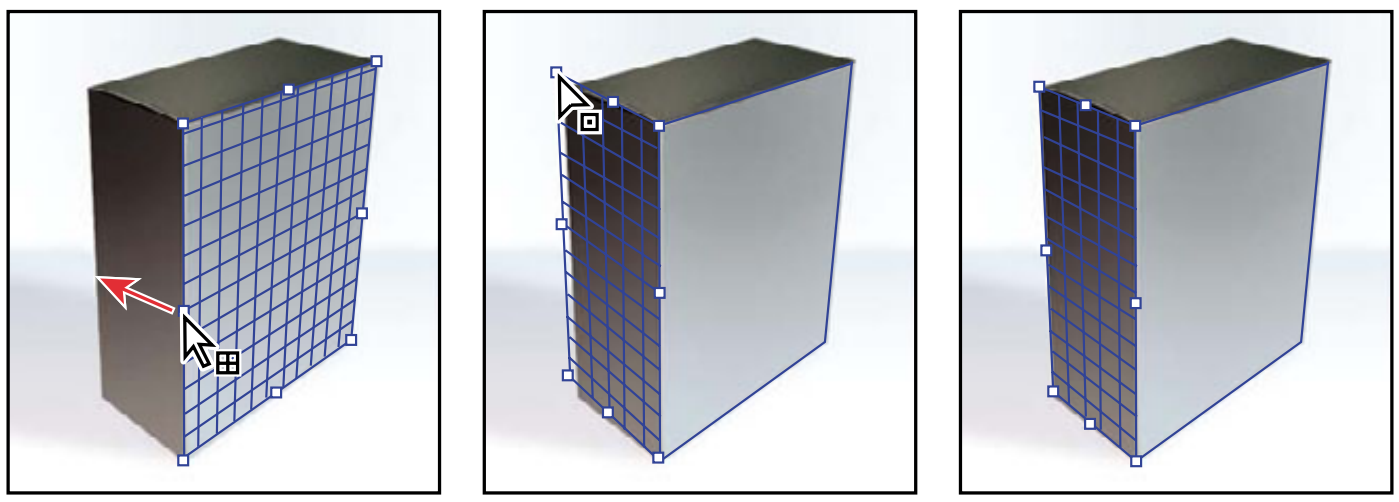


6 Use the Edit Plane pointer to adjust corner points to refine your plane, as needed.

Now you'll extend the grid to the side and top of the box to complete the perspective.

7 With the Edit Plane tool selected, press Ctrl (Windows) or Command (Mac OS), and drag the center point of the left side of the plane, extending the plane along the side of the box. As the grid appears on the side of the box, the grid on the front face disappears, but its blue border remains.

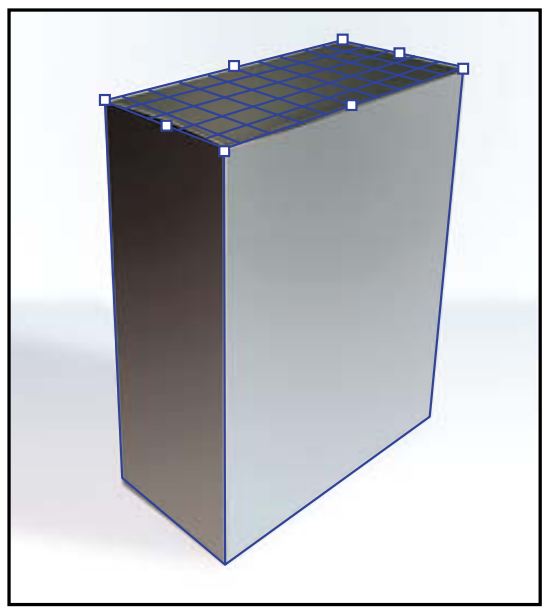
8 Use the Edit Plane pointer to adjust any corner points along the side plane, so that it more closely matches the shape of the box.



9 When you're satisfied with the grid's placement, repeat steps 7 and 8 to extend the grid across the top of the box.

If you were applying perspective to many planes, you might want to create a separate layer for each plane. Putting the Vanishing Point results in a separate layer preserves your original image and lets you use the layer opacity control, styles, and blending modes.

You're ready to add the artwork and give it perspective.



● **Note:** The final grid doesn't have to match the box dimensions exactly.

10 Press Ctrl+V (Windows) or Command+V (Mac OS) to paste the contents of the clipboard onto the grid. This action automatically selects the Marquee tool in the Vanishing Point dialog box.

11 Using the Marquee tool (⬚), select the contents, and drag it to the center of the front perspective plane so that most of the artwork appears on the front panel, but wraps around the side and top. It's important to place the artwork on the front panel so that it wraps correctly.





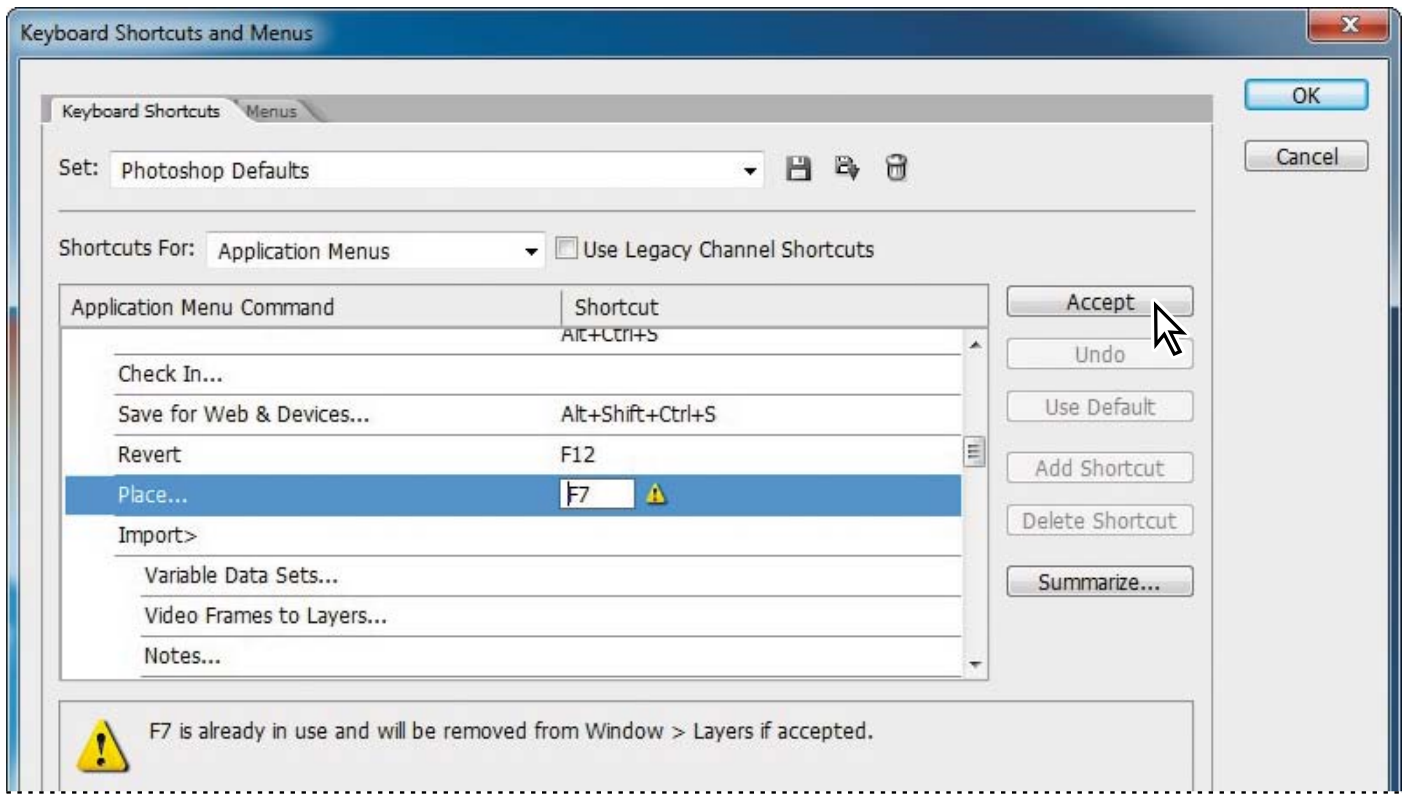
12 When you're satisfied with the positioning, click OK.

13 Choose File > Save to save your work so far.

Creating your own keyboard shortcuts

Photoshop comes with keyboard shortcuts for most common tools and commands. But you can customize shortcuts to fit your workflow. As you build the composite image, you'll place several images created in Adobe Illustrator. To make your work more efficient, create a keyboard shortcut for the Place command.

1. Choose Edit > Keyboard Shortcuts. The Keyboard Shortcuts And Menus dialog box appears.
2. Under Application Menu Command, click the triangle to the left of File to expand its contents. Scroll down to Place, and select it.
3. Press the F7 key on the keyboard to assign that key as a new shortcut. An alert appears, warning you that the F7 key is already in use. It's the keyboard shortcut for the File > Layers command, which opens the Layers panel.
4. Click Accept, and then click OK.



Placing imported artwork

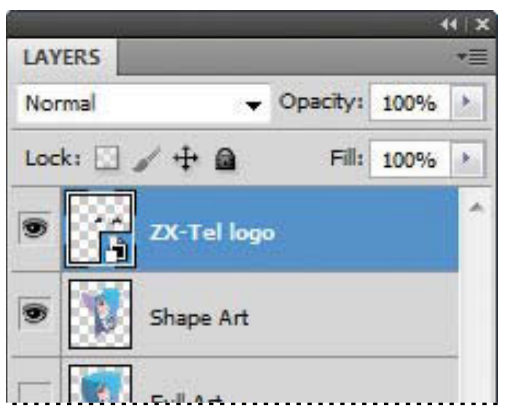
Now you'll take advantage of your new keyboard shortcut as you add more artwork to the package. The imported artwork contains the words *ZX-Tel cellular,* originally created with the Type tool in Illustrator, but then converted to a graphic. You can no longer edit the text with the Type tool. However, because of this conversion, you don't have to worry about whether others working on the file can see the type correctly if they don't have the same font installed.

1. Press F7 to open the Place dialog box.
2. Select the ZX-Tel logo.ai file in the Lesson09 folder. Click Place. The Place PDF dialog box appears.
3. Leave the settings at their defaults, and click OK to place the file. The logo image opens in Free Transform mode, so you can modify it.

The Place command adds a photo, art, or any file Photoshop supports as a Smart Object to your document. You may remember from earlier lessons that Smart Objects preserve an image's source content with all its original characteristics, enabling you to edit the Smart Object layer nondestructively. However, when you're using a Smart Object, some filters and effects are not available.

4. Drag the logo over the front panel, and then drag the corner points to resize the logo to roughly the width of the box front. Don't worry about being exact; you'll use the Vanishing Point filter to position the logo in perspective soon.
5. When you are satisfied with the positioning, press Enter or Return to place the file.





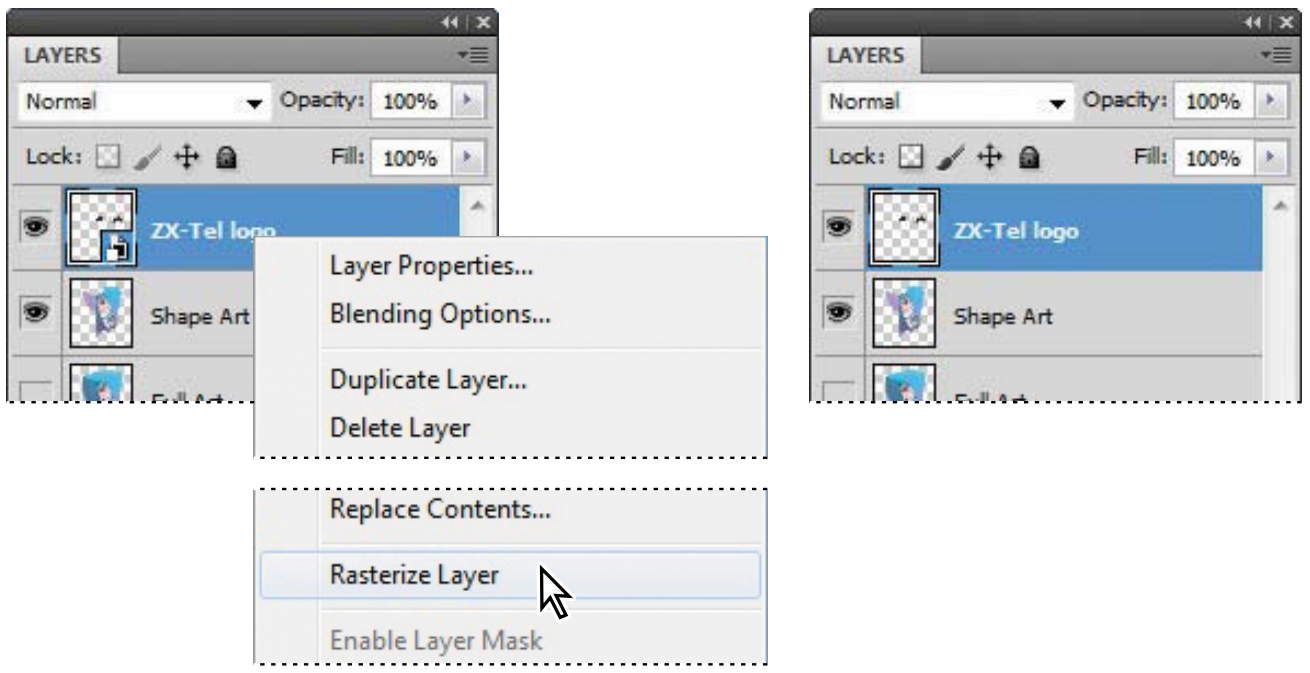
The placed image appears as the ZX-Tel logo layer at the top of the Layers panel. The icon in the lower-right corner of the layer thumbnail indicates that it is a Smart Object.

6. Choose File > Save to save your work so far.

Adding artwork in perspective

You'll apply the text you just placed to the three-dimensional box. Then, you'll transform and stylize it so that it looks realistic and in perspective. Because Smart Objects don't support the Vanishing Point filter, you'll start by converting the vector data in the Smart Object layer to pixels.

1 In the Layers panel, right-click (Windows) or Control-click (Mac OS) the ZX-Tel layer name, and choose Rasterize Layer from the context menu. This converts the Smart Object to a flat, raster layer.



2 Choose Image > Adjustments > Invert to invert the color from black to gray. This will make it easier to read the text when it's added to the box.



3 With the ZX-Tel logo layer selected in the Layers panel, choose Select > All. The entire ZX-Tel logo layer is selected.

4 Choose Edit > Cut.

5 Choose Filter > Vanishing Point to return to the perspective plane. The 3D box with the cell phone artwork is already there.

6 Press Ctrl+V (Windows) or Command+V (Mac OS) to paste the logo onto the perspective plane, and drag it into position on the front of the box.

7 Press Ctrl+T (Windows) or Command+T (Mac OS) to access the free transform handles. Drag the handles to adjust the logo so that it matches the perspective of the box. You may need to rotate the logo.

8 Press the Alt (Windows) or Option (Mac OS) key, and drag a cloned copy of the logo directly upwards and onto the top of the box. When you're satisfied with the positioning, click OK.





9 Choose File > Save to save your work.

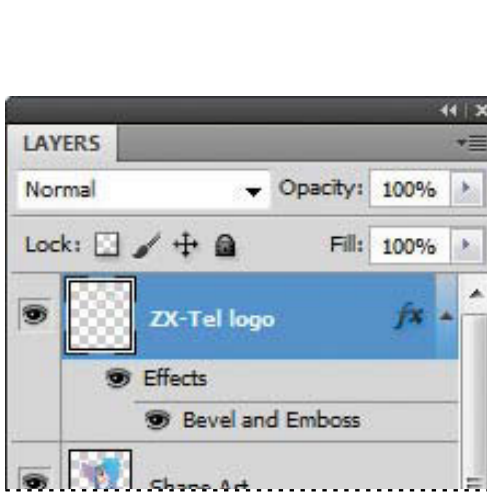
Adding a layer style

Now you'll add a layer style to give the logo some depth. Layer styles are automated effects that you can apply to a layer.

1 In the Layers panel, select the ZX-Tel logo layer.

2 Choose Layer > Layer Style > Bevel And Emboss. Leave the settings at their defaults, and click OK.





Now the logo has sharp, deep edges, to give the box the appearance of more depth.

Placing the side panel artwork

To complete the package, you'll add product copy to the side panel of the box.

1 Press F7, and select the Side Box Copy.ai file. Click Place. In the Place PDF dialog box, leave the settings at their defaults, and click OK.

2 Press the Shift key, and drag a corner to resize the placed image down to roughly the size of the side panel. Then press Enter or Return to place the artwork.



3 In the Layers panel, right-click (Windows) or Control-click (Mac OS) the side box copy layer name, and choose Rasterize Layer from the context menu.

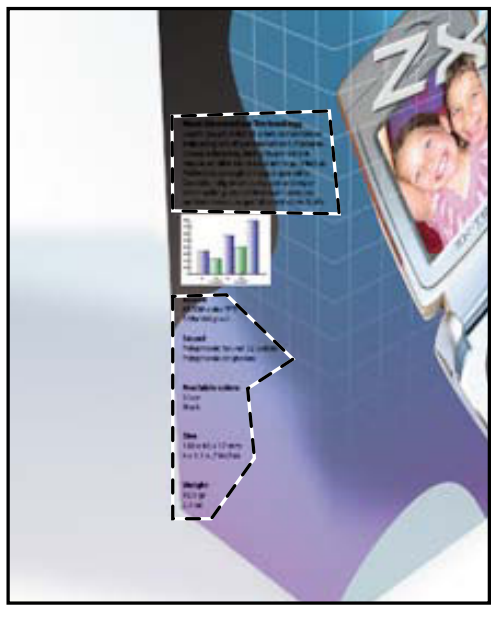
You will select the type and change its color to make it more legible.

4 Select the Polygonal Lasso tool ( ) in the Tools panel, hidden under the Lasso tool ( ).

5 Using the Polygonal Lasso tool, click at each corner to draw a box around the top block of text. Then hold down the Shift key, and draw another box around the bottom block of text to add to the selection. You don't want to include the graph.

You used the Polygonal Lasso tool because the lines of text form a slightly irregular shape. You could also use the Rectangular Marquee tool.

6 Press Ctrl+I (Windows) or Command+I (Mac OS) to invert the color from black to white. Then choose Select > Deselect.





Adding more artwork in perspective

Now you'll add the copy from the side panel to your 3D box.

1. Select the side box copy layer, and choose Select > All.
2. Press Ctrl+X (Windows) or Command+X (Mac OS) to cut the contents to the clipboard.
3. Choose Filter > Vanishing Point.
4. Press Ctrl+V (Windows) or Command+V (Mac OS) to paste the side copy artwork onto the perspective plane.
5. Position the artwork so that it fits along the side panel. If necessary, press Ctrl+T (Windows) or Command+T (Mac OS), and use the free transform handles to adjust the artwork so that it fits properly.
6. When you're satisfied with how the side copy artwork looks, click OK.



Now you'll repeat this procedure one more time to place the last piece of artwork and add it to the box in perspective.

7. Press F7, and double-click the Special Offer.ai file. Click OK to close the Place PDF dialog box. Resize the artwork to fit in the lower-left corner of the box front, and then press Enter or Return to place the file.
8. In the Layers panel, right-click (Windows) or Control-click (Mac OS) the special offer layer name, and choose Rasterize Layer from the context menu. The layer is no longer a Smart Object.

9 Place the special offer layer on the box in perspective, following the same procedure you used to add the box artwork, text, and side copy:

- With the special offer layer active, choose Select > All.
- Press Ctrl+X (Windows) or Command+X (Mac OS) to cut the contents to the clipboard.
- Choose Filter > Vanishing Point.
- Press Ctrl+V (Windows) or Command+V (Mac OS) to paste the contents from the clipboard.
- Position the artwork in the lower-left corner of the front panel, and then click OK.



10 Choose File > Save to save your work.

Adding an adjustment layer

To enhance the realism of the package, you'll add an adjustment layer that creates a shadow over the side panel.

Adjustment layers can be added to an image to apply color and tonal adjustments without permanently changing the pixel values in the image. For example, if you add a Color Balance adjustment layer to an image, you can experiment with different colors repeatedly, because the change occurs only on the adjustment layer. If you decide to return to the original pixel values, you can hide or delete the adjustment layer.

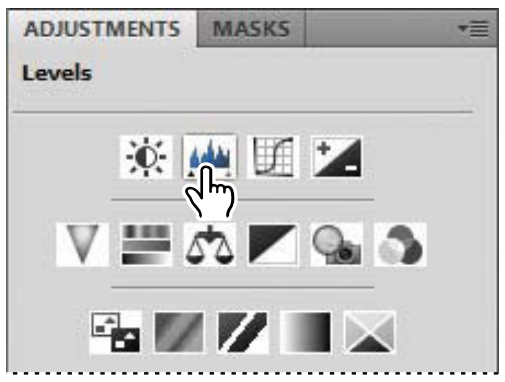
You've used adjustment layers in other lessons. Here, you'll add a Levels adjustment layer to increase the tonal range of the selection, in effect increasing the overall contrast. An adjustment layer affects all layers below it in the image's stacking order, unless a selection is active when you create it.

1 In the Layers panel, select the side box copy layer.

2 Select the Polygonal Lasso tool ( ) in the Tools panel, and click at each corner to draw a rectangular shape around the side panel.

3 Click the Levels button in the Adjustments panel to create an adjustment layer. (If the Adjustments panel isn't open, choose Window > Adjustments.)





4 In the Layers panel, double-click the name Levels 1, and rename it **Shadow**. Press Enter or Return.

5 In the Adjustments panel, drag the right (white) triangle in the Output Levels slider to about **210** to decrease the brightness. Then click the Return To Adjustment List button (◀) at the bottom of the Adjustments panel.

6 Choose File > Save to save your work.

7 Experiment by clicking the Show/Hide Visibility button (●) for the Shadow adjustment layer to see its effect. Because you made a selection before you created the adjustment layer, it affects only the selected area.When you finish, make sure that all layers but Full Art are visible.







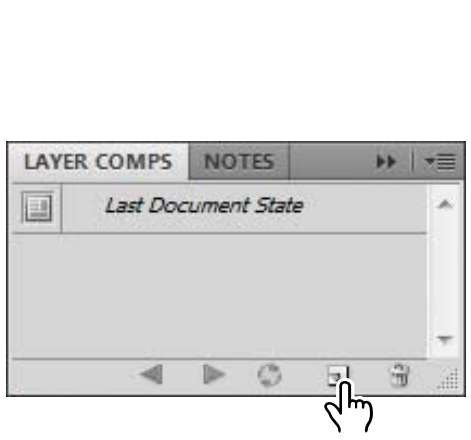


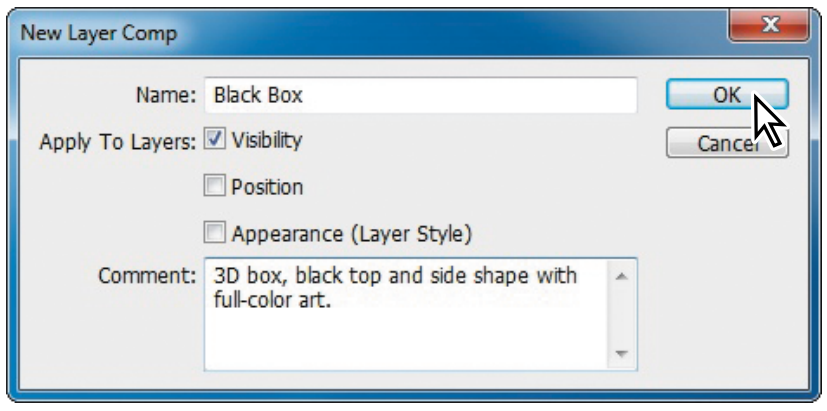
Working with layer comps

Next, you'll save this configuration as a layer comp. Layer comps let you easily switch between various combinations of layers and effects within the same Photoshop file. A layer comp is a snapshot of a state of the Layers panel.

1 Choose Window > Layer Comps.

2 At the bottom of the Layer Comps panel, click the New Layer Comp button. Name the new layer comp **Black Box**, and type a description of its appearance in the Comment box: **3D box, black top and side shape with full-color art**. Click OK.

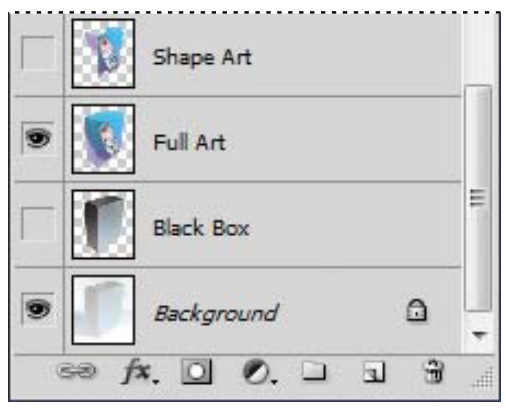




Now you'll make some changes and save the new look as a different layer comp.

3 In the Layers panel, hide the Black Box and Shape Art layers, and show the Full Art layer.

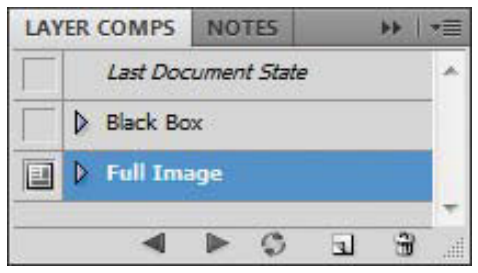


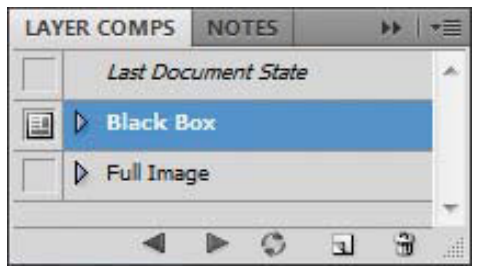


You'll save this version as a new layer comp.

4 In the Layer Comps panel, click the New Layer Comp button. Type **Full Image**, and enter a description: **3D box, blue top with full-color art**. Click OK.

5 In the Layer Comps panel, toggle the visibility icons to show and hide your two layer comps and view the differences.









You can also use layer comps to record the layer position in the document or the layer appearance, including layer styles and blending modes.

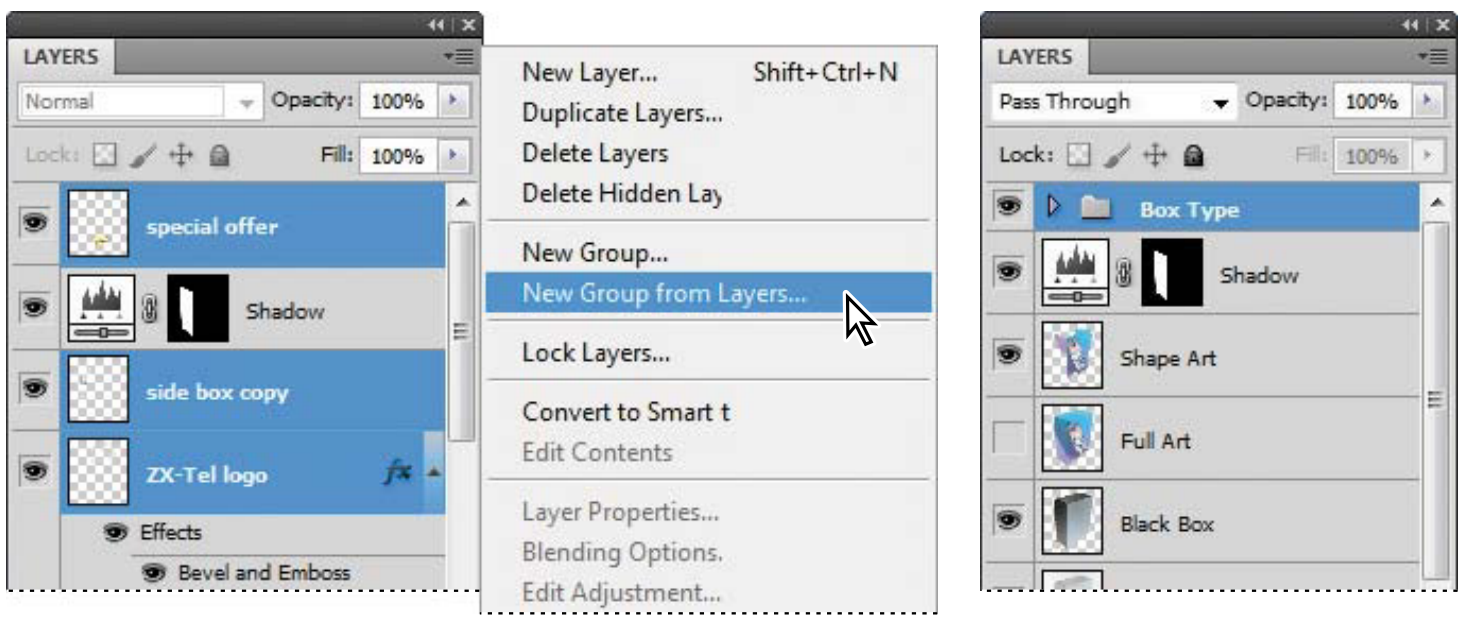
6 Choose File > Save to save your work.

Managing layers

With layer comps, you learned a great way to present different design options for a package. It's also helpful to be able to group your layers by content. You'll organize your type and art elements by creating a separate group for each.

1 In the Layers panel, Control-click (Windows) or Command-click (Mac OS) to select the special offer, side box copy, and ZX-Tel logo layers.

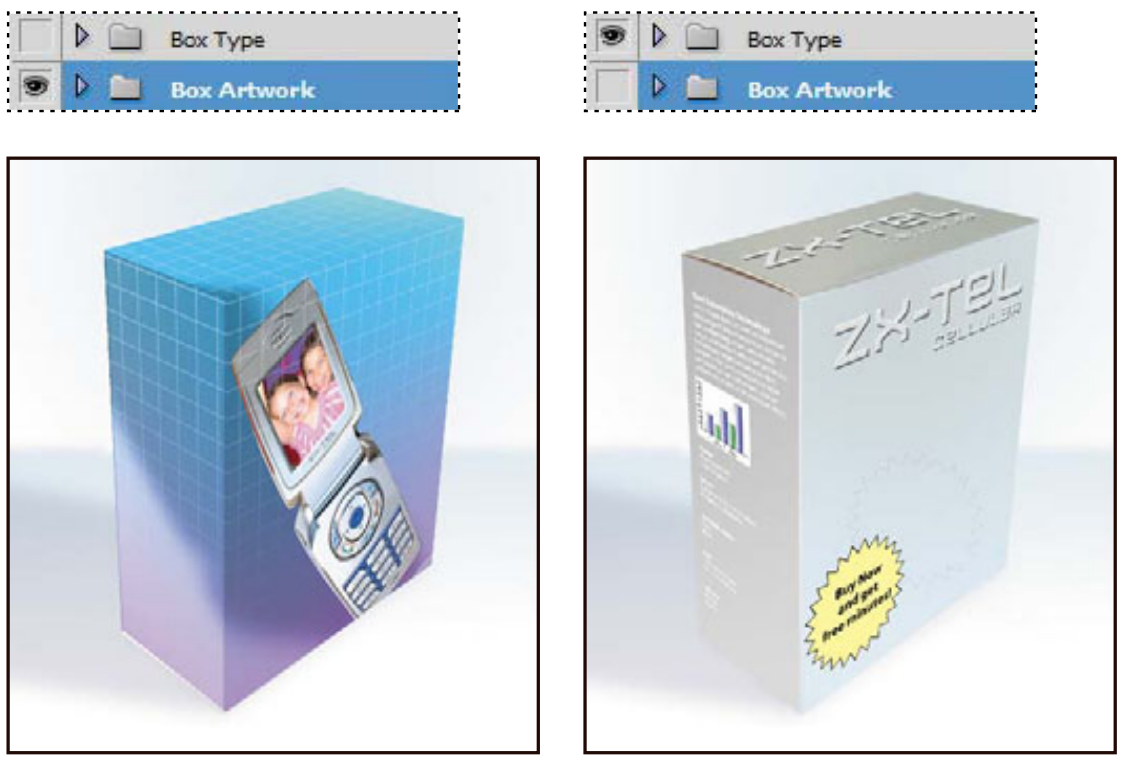
2 From the Layers panel menu, choose New Group From Layers. Type **Box Type** for the name, and click OK.



3 Select the Shadow adjustment layer, and then Shift-click the Full Art layer to select them and the layer (Shape Art) between them. Then, choose New Group From Layers from the Layers panel menu, and name this group **Box Artwork**. Then click OK.

Layer groups help you organize and manage individual layers. You can expand a layer group to view its layers, or collapse the group to simplify your view. You can change the stacking order of layers within a layer group.

4 Hide each layer group to see how the layers are grouped together. Then show the layer groups again.

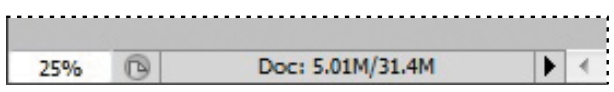


Layer groups can function like layers in a number of ways. You can select, duplicate, and move entire groups, as well as apply attributes and masks to the entire layer group. Any changes you make at the group level apply to all the layers within the group.

Flattening a layered image

As you've done in previous lessons of this book, you'll now flatten the layered image. When you flatten a file, all layers are merged into a single background, greatly reducing the size of the file. If you plan to send a file out for proofs, it's a good idea to save two versions of the file—one containing all the layers so that you can edit the file if necessary, and one flattened version to send to the print shop.

1 Note the values in the lower-left corner of the image or application window. If the display does not show the file size (such as "Doc: 5.01M/43M"), click the arrow, and choose Show > Document Sizes.

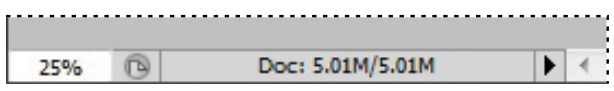


The first number is the printing size of the image, which is about the size that the saved, flattened file would be in Adobe Photoshop format. The number on the right indicates the approximate document size of the file as it is now, including layers and channels.

2 Choose Image > Duplicate, name the duplicate file **09Final.psd**, and click OK.

3 From the Layers panel menu, choose Flatten Image. Click OK when prompted to discard hidden layers.

The layers for the 09Final.psd file are combined onto a single background layer. Now the file sizes in the lower-left area of the work area are similar to the smaller number that you saw earlier. Note that flattening fills transparent areas with white.



4 Choose Edit > Undo Flatten Image.

You'll try another way to merge layers and reduce the file size.

Merging layers and layer groups

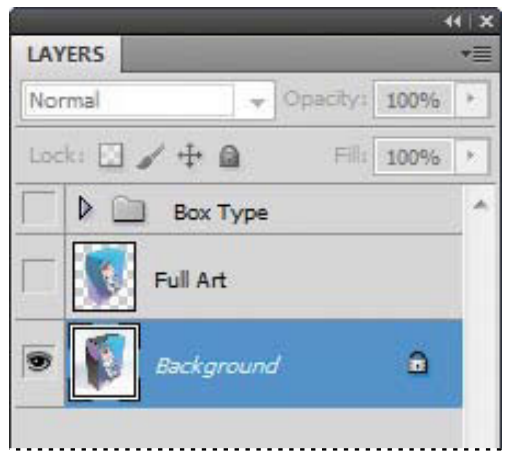
Unlike flattening a layered image, merging layers allows you to select which layers to flatten or leave unflattened.

You'll merge all the elements of the box, while keeping the Box Type layer group and Background layer untouched. This way, you could return to the file and reuse the Background and Box Type layers at any time.

1 In the Layers panel, hide the Box Type layer group to hide all of its layers.

2 Select the Box Artwork layer group in the Layers panel.

3 Choose Layer > Merge Visible. Any layers that aren't visible in the layer group will remain, unmerged, in the Layers panel.



4 Choose Edit > Undo Merge Visible.

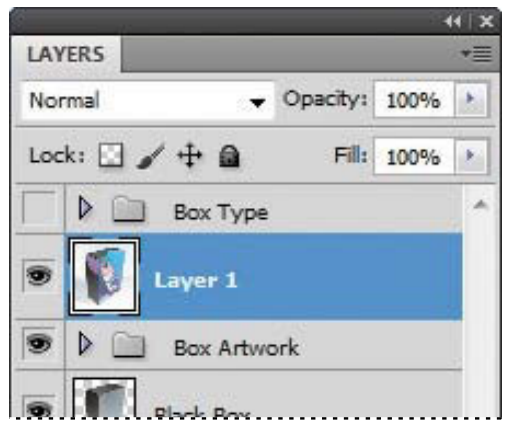
You'll try another way to merge layers and reduce the file size.

Stamping layers

You can combine the benefits of flattening an image while keeping some layers intact by *stamping* the layers. Stamping flattens two or more layers and places the flattened image into a new layer, while leaving other layers intact. This is useful if you need to use a flattened image but also need to keep some layers intact for your work.

1 In the Layers panel, select the Box Artwork group.

2 Press Alt (Windows) or Option (Mac OS) while you choose Layer > Merge Group. The Layers panel displays a new layer that includes your merged image. To reduce the file size, you could delete the original layers from the file.



3 Choose File > Save. In the Save As dialog box that appears, click Save to save the file in Photoshop format. Click OK if the Photoshop Format Options dialog box appears.

You've created a three-dimensional composite image and explored multiple ways to save final artwork.

Review questions

1 Why would you use layer groups?

2 How can you clip a layer to a shape?

3 How do adjustment layers work, and what is the benefit of using them?

4 What are layer styles, and why would you use them?

5 What is the difference between flattening, merging, and stamping layers?

Review answers

1 Layer groups let you organize and manage layers. For example, you can move all the layers in a layer group, and then apply attributes or a mask to the entire group.

2 Position the layer you want to clip above the layer you want to use as a clipping path. Select the first layer, and then press Alt or Option as you position the pointer between the layers, and click. The clipped layer is indented, with an arrow pointing down to the clipping path layer.

3 An adjustment layer applies color and tonal adjustments without changing the underlying pixels. You can show or hide, edit, delete, or mask adjustment layers without permanently affecting the image.

4 Layer styles are customizable effects that you can apply to layers. You can use them to apply changes to a layer, and you can modify or remove them at any time.

5 Flattening an image merges all layers into a single background, greatly reducing the size of the file. Merging layers lets you choose which layers to flatten; this technique combines all selected or visible layers in one layer. Stamping combines the benefits of flattening an image with the benefits of keeping some layers intact; it flattens two or more layers and places the flattened image into a new layer, while leaving other layers intact.